

Hrishikesh Deshpande

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EDUCATION

University of Illinois Urbana-Champaign

Urbana-Champaign, Illinois

B.S. Computer Engineering, GPA: 3.5/4.0

August 2023 - December 2026

Relevant Coursework: Computer Architecture, Operating Systems Design, Machine Learning Applications, Applied Parallel Programming, Autonomous Systems, Algorithms & Models of Computation, Data Structures

TECHNICAL SKILLS

Programming: Python, C, Verilog, SystemVerilog, RISC-V Assembly, C++, Java, SQL, JavaScript

Tools & Frameworks: GDB, CUDA, Git, Linux, Docker, Oscilloscope, Logic Analyzer, AMD Vivado

EXPERIENCE

NVIDIA – Systems Software Engineer Intern

May 2026 – Aug. 2026

Tegra Systems Software Team

Santa Clara, California

- Wrote ARM Trusted Firmware (EL3) C code adding CPU idle-demotion tuning knobs (shorter C-state breakeven thresholds, predictive demotion scaling) and an SMC profiler surfacing firmware idle decisions hidden from kernel
- Developed Python diagnostic tooling that captures ftrace kernel traces and power rail data from live hardware, parsing them into per-core idle-state and power telemetry and cutting hands-on time per experiment by 90%
- Automated power validation across multiple Arm server boards over SSH with persistent job queues and BMC fault recovery, driving AI-guided A/B tuning experiments improving average perf/W 5% with 10x less human effort

Symbol – Software Engineer Intern

Feb. 2026 – May 2026

Product Space UIUC

Champaign, Illinois

- Built Accounting APEX from scratch, a benchmark evaluating AI agents on real-world accounting workflows across 5 simulated company environments with 40+ interlinked financial documents each, at an RL evals startup
- Developed a full-stack World Viewer prototype with an Express proxy and the Anthropic API, dynamically generating simulated accounting environments and executing an end-to-end agentic loop of tool calls against them
- Calibrated benchmark difficulty through iterative refinement of world data, tool schemas, and rubric criteria until frontier accounting agents scored under 30%, creating evaluation headroom usable as post-training signal

Swaram – Founder & Software Developer

Aug. 2024 – Sept. 2024

Cross-Platform Music Notation App

Champaign, Illinois

- Designed and deployed a Flutter/Dart mobile app for iOS and Android with 100+ downloads and 50+ users
- Built custom music keyboards optimized for complex notation input, improving entry efficiency by 40%
- Implemented Firebase for real-time synchronization and cloud storage, enabling multi-device user experiences

PROJECTS

16 Bit Microprocessor Architecture | *SystemVerilog, Vivado, Digital Design*

Aug. 2025 – Sep. 2025

- Architected and implemented a custom 16-bit microprocessor in Vivado using SystemVerilog HDL, inspired by the Little Computer 3 (LC-3) architecture and optimized for FPGA deployment on Xilinx Spartan-7 board
- Designed datapath, control unit, ALU, register file, and memory interface with modular hardware design
- Developed a custom instruction set, assembler, and comprehensive testbench infrastructure to verify functionality and support simple programs in both simulation and physical FPGA hardware

RISC-V Operating System | *C, RISC-V Assembly, GDB*

Jan. 2025 – May 2025

- Built a Unix-style kernel from scratch with ELF loading, dynamic user stack construction, Sv39 paging, efficient trap handling, exception management, and seamless supervisor–user mode transitions for reliable execution
- Developed a VirtIO block driver with descriptor queues, interrupt completions, reset logic, and locking; implemented a writable filesystem featuring block bitmaps, page-cache dirty tracking, and hierarchical directories
- Built a preemptive threading system with context switching, lazy stack allocation, and syscalls enabling concurrent processes, inter-process pipes, and a shell supporting argument parsing and file I/O

INTERESTS

Technical: CPU/GPU Architecture, Autonomous Systems, Machine Learning, Robotics, Power/Perf

Hobbies: Cooking, Boulderling, Tennis, Photography, Tabla/Drums, Trading, Reading, Video Games